Efficient Swarm Intelligence Algorithm for Optimizing Multi-Stage Designs for Phase II Clinical Trials

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Research Background

Importance of Multi-Stage Design

- improves trial validity, reduces risk, and Enhances cost-efficiency.

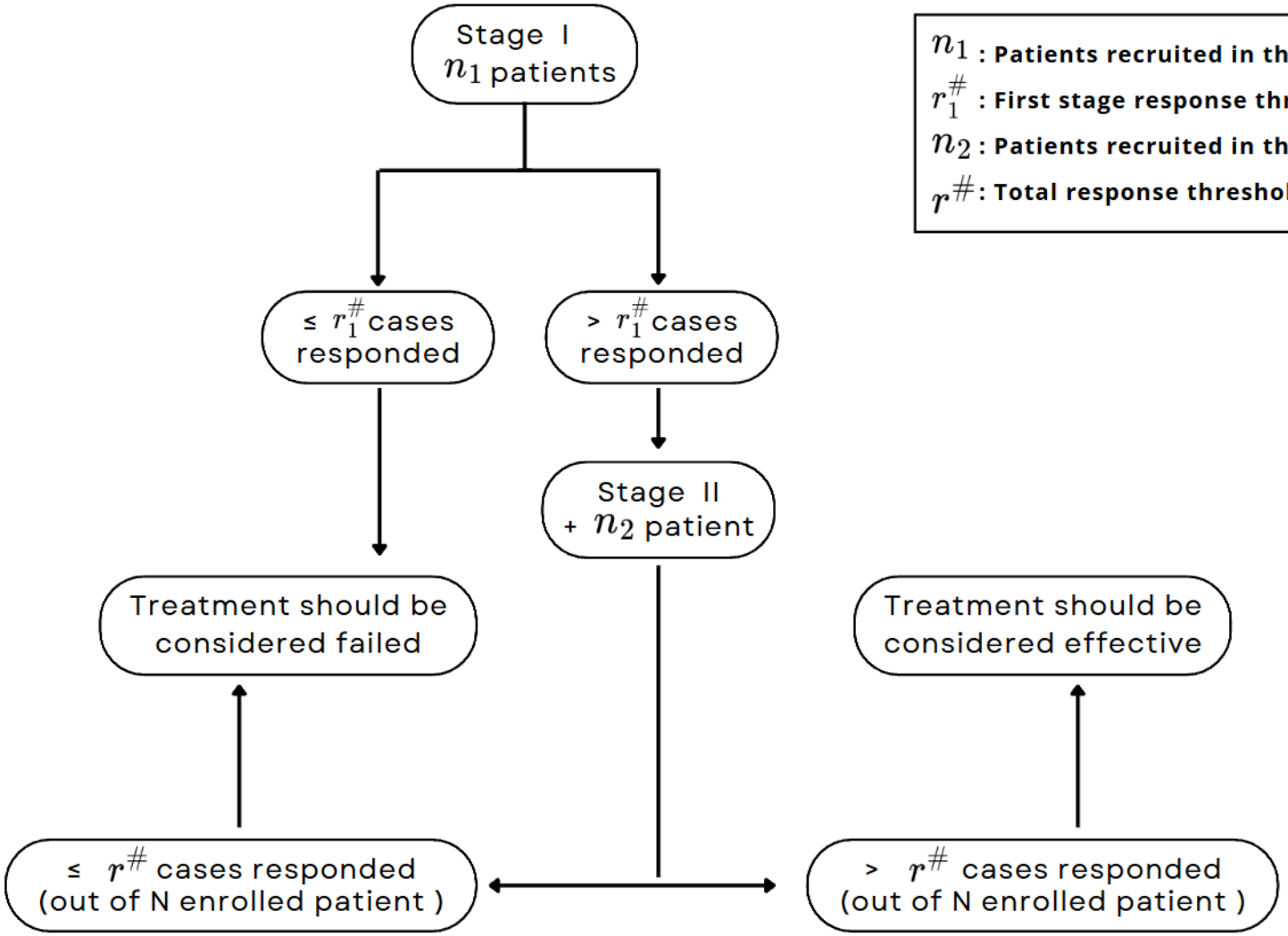
Simon's Two-Stage Design

- Flexible Evaluation: Facilitates early stopping, saving resources and minimizing patient risk (Simon, 1989).

Why Three-Stage Design?

- Optimized Sample Allocation: Ensures efficient use of sample size.
- Reduced Error Risk: Further reduces the risk of false positives and false negatives (Chen, 1997).

Literature Review: Simon's Two-Stage Design



n_1 : Patients recruited in the first stage
$r_1^{\#}$: First stage response threshold
n_2 : Patients recruited in the second stage
$r^{\#}$: Total response threshold

Literature Review: Simon's Two-Stage Design

Hypothesis Testing in Phase II Trials

- $H_0 : p \leq p_0$
- $H_1 : p \geq p_1$

p_0: Unacceptable response level
p_1: Target response level

Considerations

- Constrained by α and β
- Reducing low-efficacy treatment exposure

Probability of Early Termination at Each Stage

Binomial Distribution Formula

$$b(x; n, p) = \binom{n}{x} p^x (1-p)^{n-x}$$

- n: Total number of patients
- x: Number of successes (responses)
- p: Probability of success for each patient

Early Termination Probability (PET_1)

$$PET_1(p) = \sum_{x=0}^{r_1} b(x; n_1, p_0)$$

- n_1: Patients in the first stage
- r_1: Response threshold for the first stage

Treatment Rejection and EN (2-Stage Design)

Rejection Probability (Second Stage)

$$\sum_{x_1=r_1+1}^{\min[n_1,r_2]} b(x_1;n_1,p_0)B(r_2-x_1;n_2,p_0)$$

- n_1, n_2: Patients in stages 1 and 2
- r_1, r_2: Rejection thresholds for stages 1 and 2
- p: Success probability

Expected Sample Size (EN)

$$EN = n_1 + (1 - PET_1)n_2$$

- PET_1: Probability of early termination (stage 1)

Early Termination Probability (K-Stage Design)

Formula for $PET_k(p)$

$$PET_k(p) = \underbrace{\sum_{x_1=L_1}^{U_1} \sum_{x_2=L_2}^{U_2} \cdots \sum_{x_{k-1}=L_{k-1}}^{U_{k-1}}}_{k-1 \text{ Summations}} \left[\prod_{s=1}^{k-1} b(x_s; n_s, p_0)\right] B\left(r_k = \sum_{s=1}^{k-1} x_s, n_k, p_0\right)$$

- $b(x_s; n_s, p)$: Binomial probability at stage s.
- L_j, U_j: Lower and upper bounds of stage j.

Formula for Expected Sample Size (EN)

$$EN = n_1 + (1 - PET_1)n_2 + \cdots + (1 - PET_{k-2})n_{k-1} + (1 - PET_{k-1})n_k$$

- (EN_{p_0}) is the primary objective function of the PSO algorithm.

Total Combination Possibilities

p_0	p_1	Stage 1	Stage 2	EN(p_0)	PET$_1$(p_0)
0.1	0.3	1/12	5/35	19.8	0.65

- **2-Stage Design:** $1,105,524$ possibilities
- **3-Stage Design:** $213,672,935$ possibilities
- **4-Stage Design:** $22,546,940,047$ possibilities
- **5-Stage Design:** $1,530,973,869,444$ possibilities

Note: Enumeration range:
Total sample size $N = n_1 + \ldots + n_k$, where $30 \leq N \leq 70$.

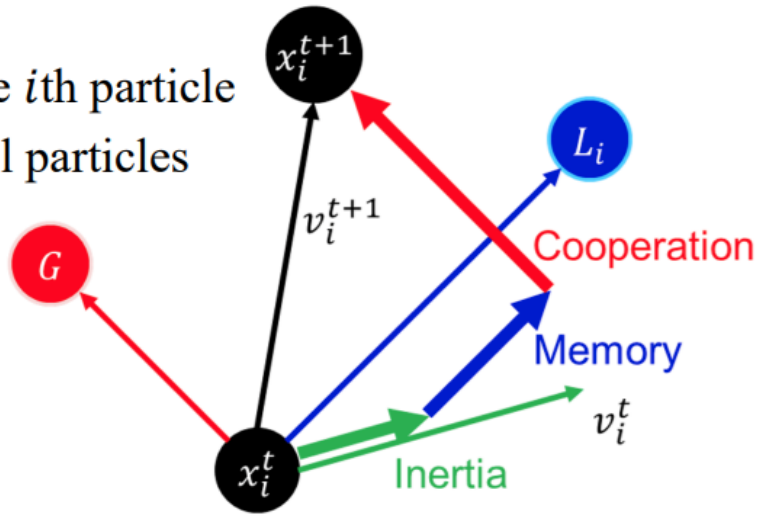
PSO(Particle Swarm Optimization)

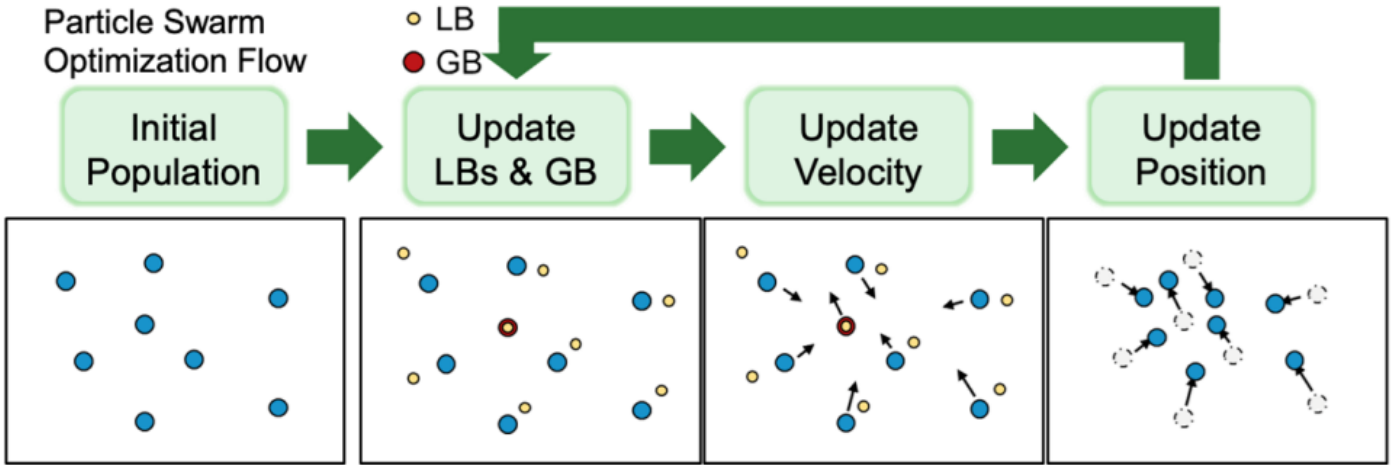
- Local Best L_i: historical best solution of the ith particle
- Global Best G: historical best solution of all particles

$$v_i^{t+1} = w \cdot v_i^t + c_1 \cdot r_1(L_i - x_i^t) + c_2 \cdot r_2(G - x_i^t)$$

$$x_i^{t+1} = x_i^t + v_i^{t+1}$$

- w, c_1, c_2 are tuning parameters.
- r_1, r_2 are uniform random vectors.





3-stage Optimal Design

p_0	p_1	Stage 1	Stage 2	Stage 3	EN(p_0)	$PET_1(p_0)$	$PET_2(p_0)$
0.05	0.25	0/9	1/18	2/26	13.78	0.63	0.82
0.10	0.30	0/10	2/19	4/26	17.79	0.35	0.72
0.15	0.35	1/12	3/21	7/33	21.11	0.44	0.66
0.20	0.40	1/10	6/26	11/43	23.86	0.38	0.77
0.25	0.45	3/16	7/25	13/41	25.55	0.41	0.74
0.30	0.50	3/13	9/28	17/46	26.75	0.42	0.7
0.35	0.55	6/18	13/33	20/48	27.84	0.55	0.8
0.40	0.60	6/17	12/28	21/44	27.59	0.45	0.72
0.45	0.65	5/13	13/27	26/50	27.22	0.43	0.73
0.50	0.70	4/10	13/25	27/47	26.08	0.38	0.69
0.55	0.75	5/11	12/21	27/43	24.36	0.37	0.7
0.60	0.80	6/11	14/22	29/43	22.30	0.47	0.74
0.65	0.85	5/9	13/19	25/34	19.23	0.39	0.72
0.70	0.90	5/8	11/15	22/28	15.45	0.45	0.72
0.75	0.95	3/5	6/8	16/19	10.94	0.37	0.63
$p_1 - p_0 = 0.20$ and $\alpha = 0.10$, $\beta = 0.10$							

Chen 1997 vs. PSO Results

Improved Combinations for 3-Stage Designs

P_0	P_1		Stage 1	Stage 2	Stage 3	EN(p_0)	PET1(p_0)	PET2(p_0)
$\alpha = 0.10$, $\beta = 0.10$								
0.40	0.60	Chen	7/18	9/26	22/46	29.81	0.56	0.58
		PSO	6/17	12/28	21/44	27.59	0.45	0.72
0.50	0.70	Chen	4/10	12/24	26/45	26.37	0.38	0.64
		PSO	4/10	13/25	27/47	26.08	0.38	0.69
$\alpha = 0.05$, $\beta = 0.20$								
0.40	0.60	Chen	3/9	10/23	23/46	21.74	0.48	0.76
		PSO	3/9	11/24	25/51	21.71	0.48	0.82

4-stage Optimal Design

p_0	p_1	Stage 1	Stage 2	Stage 3	Stage 4	EN(p_0)	$PET_1(p_0)$	$PET_2(p_0)$	$PET_3(p_0)$
0.05	0.25	0/9	0/17	1/18	2/26	13.78	0.63	0.63	0.82
0.10	0.30	0/11	1/15	2/19	4/26	17.37	0.31	0.57	0.73
0.15	0.35	0/9	2/16	4/26	7/33	20.55	0.23	0.58	0.72
0.20	0.40	1/10	4/21	7/30	11/43	22.73	0.38	0.63	0.80
0.25	0.45	1/10	4/18	8/28	14/45	24.42	0.24	0.54	0.78
0.30	0.50	2/11	6/21	11/33	17/46	25.77	0.31	0.59	0.78
0.35	0.55	4/14	10/27	15/38	20/48	26.55	0.42	0.70	0.82
0.40	0.60	4/13	10/24	16/36	23/49	26.29	0.30	0.62	0.75
0.45	0.65	3/10	9/20	16/33	25/48	25.81	0.27	0.61	0.77
0.50	0.70	4/11	8/17	15/28	26/45	24.96	0.27	0.52	0.75
0.55	0.75	5/11	10/18	19/32	28/45	23.20	0.37	0.63	0.76
0.60	0.80	5/10	9/15	17/27	27/40	20.91	0.37	0.61	0.76
0.65	0.85	5/9	13/19	20/28	24/33	18.32	0.39	0.72	0.75
0.70	0.90	1/3	6/9	15/20	24/31	14.72	0.22	0.56	0.80
0.75	0.95	3/5	6/8	11/14	16/19	10.23	0.37	0.63	0.77
$p_1 - p_0 = 0.20$ and $\alpha = 0.10$, $\beta = 0.10$									

EN(p_0) Comparison between 3 & 4-stage Designs

P_0	P_1	EN(p_0) (Optimal)		Variation
		3-stage	4-stage	
0.05	0.25	13.78	13.78	0
0.10	0.30	17.79	17.37	-0.42
0.15	0.35	21.11	20.55	-0.56
0.20	0.40	23.86	22.73	-1.13
0.25	0.45	25.55	24.42	-1.13
0.30	0.50	26.75	25.77	-0.98
0.35	0.55	27.84	26.55	-1.29
0.40	0.60	29.81	26.29	-3.52
0.45	0.65	27.22	25.81	-1.41
0.50	0.70	26.37	24.96	-1.41
0.55	0.75	24.36	23.20	-1.16
0.60	0.80	22.30	20.91	-1.39
0.65	0.85	19.23	18.32	-0.91
0.70	0.90	15.45	14.72	-0.73
0.75	0.95	10.94	10.23	-0.71

Challenges, and Future Plans

- **Challenges**:
  - Stability issues, convergence speed, and sensitivity to parameters .
  - Inconsistent results across multiple runs.
- **Future Plans**:
  - Enhancing algorithm stability and convergence efficiency.
  - Integrating **minimax design** into the algorithm.
  - Adapting the Approach to Broader Clinical Trial Scenarios (e.g., Platform Trials).

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